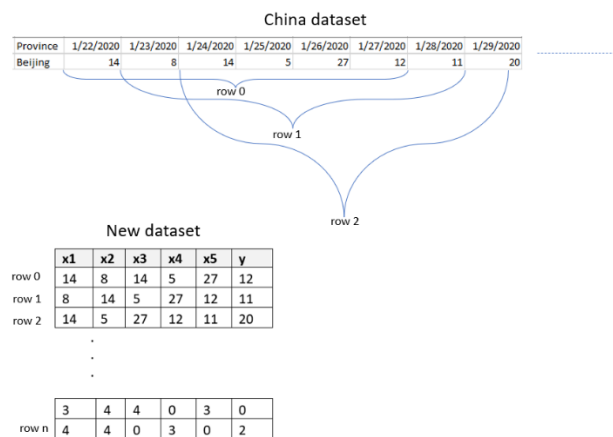


Bishop's University
Department of Computer Science
Intelligent systems and Neural Nets – CS417
FL2023 – Project

A dataset about the daily confirmed covid-19 cases in China is provided to you (*daily_confirmed_cases.csv*).

1. **Data preparation:** select the province Beijing (all data about it) then create a new dataset according to the following process:
 - 1) Iteratively go through all daily cases from the first day to the most recent 6th day. At each iteration slice five cases to use as input and the 6th case to use as output. This allows you to create the new dataset. This is illustrated in the figure below. The column names **x1** to **x5** refer to the input (predictors) and the last column named “**y**” refers to the output (6th case).



- 2) Save the obtained dataframe into a **csv** file under the name “Beijing_covid19_regression.csv”
2. **Regression problem:** Let's treat the problem as a regression problem and the question is: if we observe the confirmed cases for the last five days (**x1** to **x5**), can we predict the confirmed case for the 6th day (**y**)?
 - 1) Divide the new dataset into training set and test set.
 - 2) Train a simple linear regression model on the training set and test it on the test set.
 - 3) Compute the **error rate**.
 - 4) Repeat 2) and 3) for LASSO regression model.
 - 5) Analyze the LASSO with respect to the parameter ‘**alpha**’. Plot the change of the **error rate** for ‘**alpha**’ in [0, 1, 10, 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000].
3. **Classification problem:**
 - 1) Assume we want to classify whether the next confirmed case (6th for example) as **goup** if it will go up, **godown** if it will go down or **stable** if it doesn't change given that we observed the previous five confirmed cases.
From the previous dataset, “Beijing_covid19_regression.csv”, you will create a new dataset with a new column holding the class of each observation. This is the procedure:
 - Create a copy of the dataset “Beijing_covid19_regression.csv”, except that the last column will hold:
 - **goup** if $x_6 - x_5 > 0$.
 - **godown** if $x_6 - x_5 < 0$.
 - **stable** if $x_6 - x_5 = 0$.

- The procedure is illustrated in the figure below.

x1	x2	x3	x4	x5	x6		x6-x5	Label
14	8	14	5	27	12		-15	godown
8	14	5	27	12	11		-1	godown
14	5	27	12	11	20		9	goup
⋮								
3	4	4	0	3	0		-3	godown
4	4	0	3	0	2		2	goup

- The new dataset should look like this below. Save it into a csv file named “Beijing_covid19_classification.csv”.

x1	x2	x3	x4	x5	Label
14	8	14	5	27	godown
8	14	5	27	12	godown
14	5	27	12	11	goup
⋮					
3	4	4	0	3	godown
4	4	0	3	0	goup

- 2) Apply logistic regression (LR) to solve the problem of classification.
- 3) Analyze the performance of LR with respect to the parameters: ‘*penalty*’ and ‘*C*’.
 - a. Report the results in a Table.
 - b. Plot the results as graphs.
 - c. Provide a brief discussion of the finding.

4. Clustering problem:

Hypothesis: Nearby cities may show similar covid cases.

For this part you will use two datasets:

- *daily_confirmed_cases.csv*: a dataset for daily confirmed cases.
- *coordinates.csv*: longitude and latitude coordinates of China’s cities.

- 1) Cluster China’s cities based on the confirmed cases using *daily_confirmed_cases.csv*.
- 2) Cluster China’s cities based on the coordinates (i.e., based on closeness) using *coordinates.csv*.
- 3) Analyze the clustering results in 1) and 2) to validate the hypothesis stated above.

Note: you may use a simple k-means algorithm.